20. If R = 5000 units / year, C_1= 20 paise, C_3= ₹ 20, then EOQ is

(a) 5000 (b) 100 (c) 1000 (d) 200

Miscellaneous Problems

1. The total cost function for the production of x units of an item is given by $C = 10 - 4x^3 + 3x^4$ find the (i) average cost function (ii) marginal cost function (iii) marginal average cost fuction.
2. Find out the indicated elasticity for the following functions

   (i) $p = xe^x$, $x > 0$; η_s

   (ii) $p = 10e^{-\frac{x}{3}}$, $x > 0$; η_d
3. Find the elasticity of supply when the supply function is given by $x = 2p^2 + 5$ at p=1.
4. For the demand function $p = 100 - 6x^2$, find the marginal revenue and also show that $MR = p\left[1 - \frac{1}{\eta_d}\right]$
5. The total cost function y for x units is given by $y = 4x\left(\frac{x+2}{x+1}\right) + 6$. Prove that marginal cost[MC] decreases as x increases.
6. For the cost function $C = 2000 + 1800x - 75x^2 + x^3$ find when the total cost (C) is increasing and when it is decreasing.
7. A certain manufacturing concern has total cost function $C = 15 + 9x - 6x^2 + x^3$. Find x, when the total cost is minimum.
8. Let $u = \log\frac{x^4 - y^4}{x - y}$. Using Euler's theorem show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$.
9. Verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for $u = x^3 + 3x^2y^2 + y^3$
10. If $f(x, y) = 3x^2 + 4y^3 + 6xy - x^2y^3 + 7$, then show that $f_{yy}(1,1) = 18$.

Summary

- Demand is the relationship between the quantity demanded and the price of a commodity.
- Supply is the relationship between the quantity supplied and the price of a commodity.
- Cost is the amount spent on the production of a commodity.
- Revenue is the amount realised by selling the output produced on commodity.
- Profit is the excess of total revenue over the cost of production.
- Elasticity of a function $y = f(x)$ at a point x is the limiting case of ratio of the relative change in y to the relative change in x
- Equilibrium price is the price at which the demand of a commodity is equal to its supply.
- Marginal cost is interpreted as the approximate change in production cost of $(x+1)^{th}$ unit, when the production level is x units.

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